Digital Logic Mini Project

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Project 2
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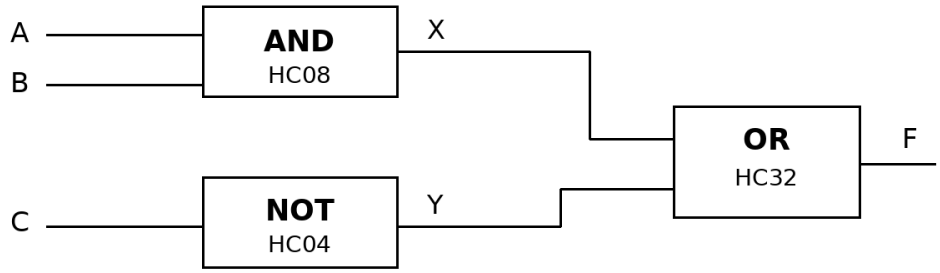
Goal

I built a circuit with three digital inputs and checked how its outputs changed. I used one AND gate, one NOT gate and one OR gate. All eight input combinations matched the expected LED states in my final test log. I also tested an open LED connection and soldered a separate LED circuit.

Logic design

X = A AND B; Y = NOT C; F = X OR Y

F is HIGH when C is LOW, or when A and B are both HIGH. Each output has its own red LED. A lit output LED represents HIGH.



Signal flow through the three logic gates.

Parts and setup

I placed the ICs across the breadboard gap. Each uses pin 14 for +5 V, pin 7 for GND and a nearby 100 nF capacitor. The three switch inputs have 10k pull-down resistors, and each LED has a 1k series resistor. Unused inputs connect to GND; unused outputs and the fourth switch position remain open.

The DFR0140 module supplies regulated 5 V from a 9 V adapter. On the board, Y is closest to the power module, X is in the middle and F is farthest away.

Readings

I used the Klein MM420 in DC voltage mode. For each input combination, I observed all three LEDs and measured each output relative to GND. X was measured at HC08 pin 3, Y at HC04 pin 2 and F at HC32 pin 3. The black probe used pin 7 of the corresponding IC.

Supply measurement	**Voltage**
HC04 pin 14 relative to pin 7	4.98 V
HC08 pin 14 relative to pin 7	4.99 V
HC32 pin 14 relative to pin 7	4.98 V

In the table, 0 means OFF or LOW and 1 means ON or HIGH. Expected and observed LED states are ordered X, Y, F.

Test	**ABC**	**Expected XYF**	**X V**	**Y V**	**F V**	**LED XYF**	**Result**
1	000	011	0	4.53	4.47	011	Pass
2	001	000	0	0	0	000	Pass
3	010	011	0	4.52	4.40	011	Pass
4	011	000	0	0	0	000	Pass
5	100	011	0	4.45	4.53	011	Pass
6	101	000	0	0	0	000	Pass
7	110	111	4.43	4.56	4.54	111	Pass
8	111	101	4.49	0	4.51	101	Pass

Results

All eight recorded LED patterns matched the truth table. The HIGH output readings ranged from 4.40 to 4.56 V. LOW readings were recorded as 0 V; separate millivolt readings were not logged. The final 111 result has Y = 0 V, as expected for NOT C.

An IC output driving an LED is loaded, so its HIGH voltage can be lower than the supply. These measurements show the static logic behavior. They do not measure switching speed or establish the accuracy of the multimeter.

Separate HIGH and LOW input-voltage readings for A, B and C were not recorded in the final data table. The recorded supply measurements are at the IC pins, rather than separate measurements of both power rails.

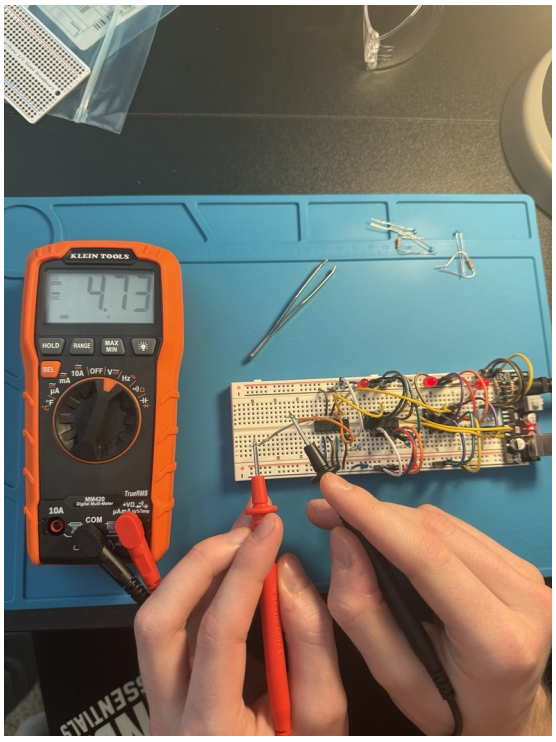
Open connection test

Expected behavior: LED F should light when the OR output is HIGH. I switched power off and disconnected the connection between the OR output and its LED branch. After applying power, LED F stayed off.

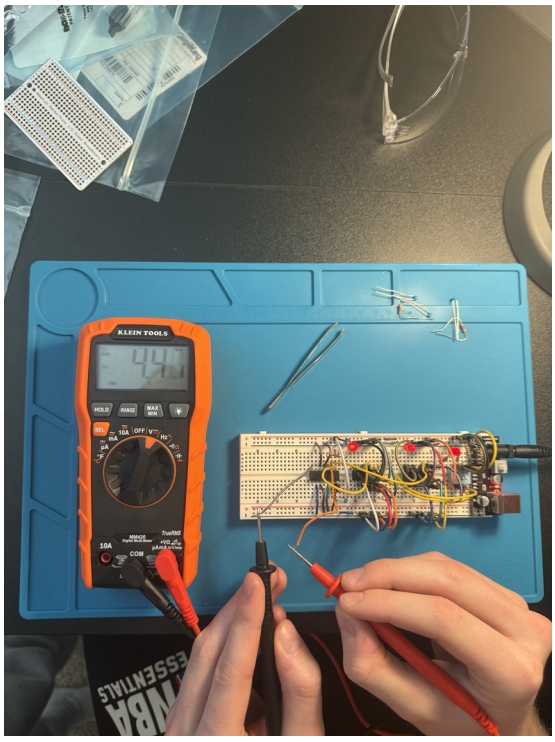
I measured the OR output relative to GND. A HIGH reading was still present. This showed that the indicator branch could be open even while the gate was producing a HIGH signal. I switched power off, restored the connection and checked that LED F lit again.

Condition	F voltage	LED F
Open branch, earlier written reading	4.52 V	OFF
Open branch, photographed reading	4.73 V	OFF
Connection restored, photographed reading	4.47 V	ON

The earlier written value and the photographed value are separate readings. The input setting was not confirmed for the troubleshooting photo pair, so these readings are kept separate from the eight-case table.



Open branch. The meter shows 4.73 V while LED F is off.



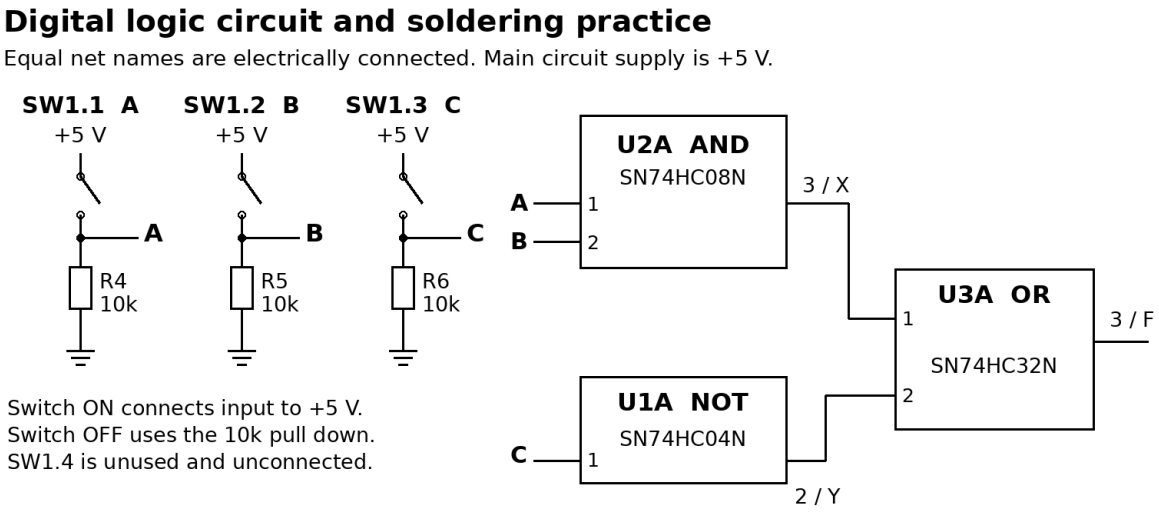
Restored branch. The meter shows 4.47 V and LED F is on.

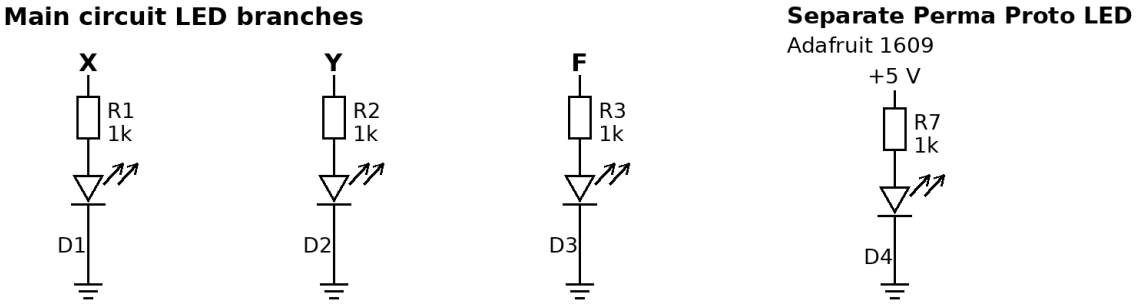
What I found

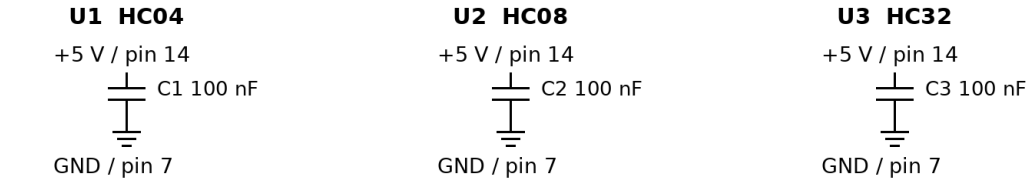
The LED branch had an open connection. Restoring that connection allowed current to flow through the LED and its series resistor. I learned to check the signal at the IC output before deciding that an unlit LED means the gate has failed.

Circuit diagram

This schematic follows the recorded electrical connections. Named signal nets connect the switches, gates and LED branches. U1, U2 and U3 refer to the NOT, AND and OR ICs. It is not a breadboard hole-location drawing.







Unused pins

U1 HC04	U2 HC08	U3 HC32
GND: 3, 5, 9, 11, 13	GND: 4, 5, 9, 10, 12, 13	GND: 4, 5, 9, 10, 12, 13
Open: 4, 6, 8, 10, 12	Open: 6, 8, 11	Open: 6, 8, 11

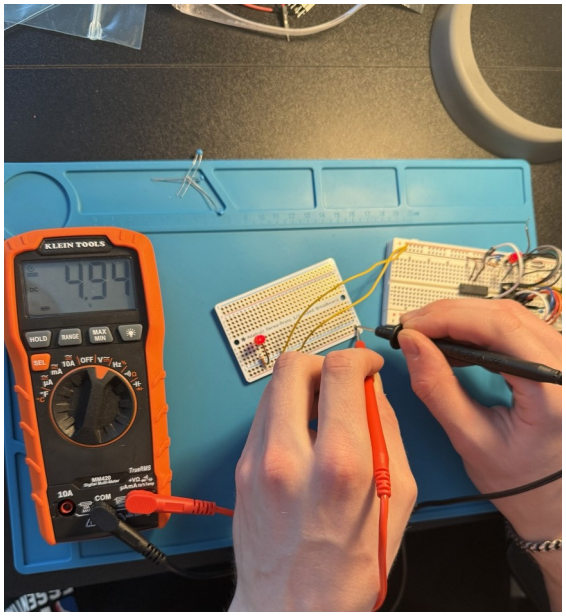
Main logic circuit, power connections, unused pins and separate Perma-Proto LED circuit. D1 = X, D2 = Y, D3 = F, D4 = soldering practice.

Soldering practice

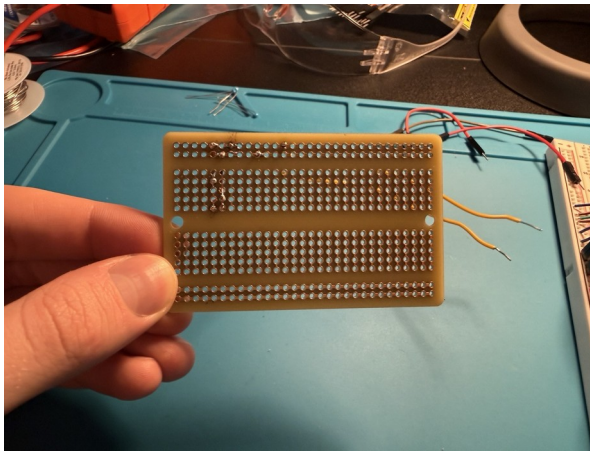
I kept the three-IC circuit on the solderless breadboard. For soldering practice, I built a separate circuit on Adafruit 1609 Perma-Proto: +5 V through a 1k resistor, then a red LED, then GND.

I used a Hakko FX-888DX station and Chip Quik RASWLF.031 lead-free solder. The starting iron setting was 350 °C. I worked on a balcony with a silicone mat and safety glasses. I soldered the resistor, LED, ground connection and two power wires.

With power disconnected, I checked continuity from the LED cathode to GND and checked that the positive and ground rails did not give a continuity beep. The powered LED lit. The photographed supply measurement at the Perma-Proto rails is 4.94 V DC.



The working soldered LED circuit with 4.94 V measured at its rails.



The underside of the practice board and its solder joints.

What I would improve

Some joints have excess solder and uneven shapes. The power wires also need clear polarity labels and better strain relief for repeated handling. Next time I would use less solder, keep more consistent heating time and keep the original male contacts on the breadboard ends of the power leads.

What I learned

I assembled and tested a small digital logic circuit, checked all eight input combinations and diagnosed an open LED branch using a multimeter. I also completed a working soldered LED circuit. This helped me connect a truth table to real voltages, wiring and indicator behavior.

Bill of materials and references

Counts below cover installed components. Purchased spare ICs, LEDs, resistors and capacitors are excluded. The adapter and power module were shared between the main circuit and the soldering test.

Component and model	Main	Soldered
SN74HC04N: NOT IC, PDIP-14	1	0
SN74HC08N: AND IC, PDIP-14	1	0
SN74HC32N: OR IC, PDIP-14	1	0
CTS 206-4: 4-position DIP switch; 3 positions used	1	0
Kingbright WP7113ID: Red 5 mm LED	3	1
Yageo CFR-25JB-52-1K: 1 kohm series resistor	3	1
Yageo CFR-25JB-52-10K: 10 kohm input pull-down resistor	3	0
Murata RHEL81H104K0A2H03B: 100 nF decoupling capacitor	3	0
Adafruit 239: 830-point solderless breadboard	1	0
Adafruit 1609: Half-sized Perma-Proto	0	1
DFRobot DFR0140: Shared regulated 5 V supply	1	0
GlobTek WR9HD1000CCP-F(R6B): Shared 9 V DC input adapter	1	0
Jumper wire and lead offcuts: Installed count not recorded; practice board uses 2 power leads	As used	As used

Tools used: Klein MM420 multimeter, Hakko FX-888DX station and holder, Chip Quik RASWLF.031 solder, flush cutters, safety glasses and silicone mat. No microcontroller or MATLAB program is required for this circuit.

Manufacturer references

Texas Instruments SN74HC04 datasheet
https://www.ti.com/lit/ds/symlink/sn74hc04.pdf

Texas Instruments SN74HC08 datasheet
https://www.ti.com/lit/ds/symlink/sn74hc08.pdf

Texas Instruments SN74HC32 datasheet
https://www.ti.com/lit/ds/symlink/sn74hc32.pdf

DFRobot DFR0140 schematic
https://dfimg.dfrobot.com/enshop/image/data/DFR0140/DFR0140_Schematics.pdf

Adafruit Perma-Proto 1609
https://www.adafruit.com/product/1609

Kingbright WP7113ID datasheet
https://www.kingbrightusa.com/images/catalog/SPEC/WP7113ID.pdf

Adafruit soldering guide
https://learn.adafruit.com/adafruit-guide-excellent-soldering/making-a-good-solder-joint